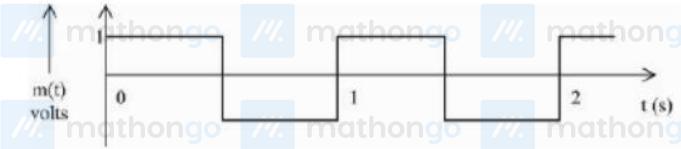


Q1 - 24 January - Shift 1

A modulating signal is a square wave, as shown in the figure.



If the carrier wave is given as $c(t) = 2 \sin(8\pi t)$ volts, the modulation index is :

- (1) $1/4$ (2) 1
(3) $1/3$ (4) $1/2$

Space for your notes:

Q2 - 24 January - Shift 2

Match List I with List II

LIST I		LIST II	
A.	AM Broadcast	I.	88-108 MHz
B.	FM Broadcast	II.	540-1600 kHz
C.	Television	III.	3.7-4.2 GHz
D.	Satellite Communication	IV.	54MHz - 590MHz

Choose the correct answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
(2) A-IV, B-III, C-I, D-II
(3) A-II, B-III, C-I, D-IV
(4) A-I, B-III, C-II, D-IV

Space for your notes:

Q3 - 25 January - Shift 1

A message signal of frequency 5 kHz is used to modulate a carrier signal of frequency 2 MHz. The bandwidth for amplitude modulation is:

- (1) 5 kHz (2) 20 kHz
(3) 10 kHz (4) 2.5 kHz

Space for your notes:

Q4 - 29 January - Shift 1

If the height of transmitting and receiving antennas are 80 m each, the maximum line of sight distance will be:

Space for your notes:

Given : Earth's radius = 6.4×10^6 m.

- (1) 32 km (2) 28 km
(3) 36 km (4) 64 km

Q5 - 29 January - Shift 2

The modulation index for an A.M. wave having maximum and minimum peak to peak voltages of 14 mV and 6 mV respectively is :

Space for your notes:

- (1) 1.4 (2) 0.4
(3) 0.2 (4) 0.6

Q6 - 30 January - Shift 2

Match List I with List II :

	List I		List II
A.	Attenuation	I	Combination of a receiver and transmitter.
B.	Transducer	II	Process of retrieval of information from the carrier wave at received
C.	Demodulation	III	Converts one form of energy into another
D.	Repeater	IV	Loss of strength of a signal while propagating through a medium

Space for your notes:

Choose the correct answer from the options given below :

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-IV, B-III, C-I, D-II
- (4) A-IV, B-III, C-II, D-I

Q7 - 31 January - Shift 1

The amplitude of $15\sin(1000\pi t)$ is modulated by $10\sin(4\pi t)$ signal. The amplitude modulated signal contains frequency(ies) of

- (A) 500 Hz
- (B) 2 Hz
- (C) 250 Hz
- (D) 498 Hz
- (E) 502 Hz

Choose the correct answer from the options given below:

- (1) A only
- (2) A, D and E only
- (3) B only
- (4) A and B only

Space for your notes:

Q8 - 31 January - Shift 2

Given below are two statements

Statement I: For transmitting a signal, size of antenna (l) should be comparable to wavelength of signal (at least $l = \frac{\lambda}{4}$ in dimension).

Statement II: In amplitude modulation, amplitude of carrier wave remains constant (unchanged).

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both **Statement I** and **Statement II** are correct
- (2) Both **Statement I** and **Statement II** are incorrect
- (3) **Statement I** is incorrect but **Statement II** is correct
- (4) **Statement I** is correct but **Statement II** is incorrect

Space for your notes:

Q9 - 01 February - Shift 1

Which of the following frequencies does not belong to FM broadcast.

- | | |
|-------------|------------|
| (1) 106 MHz | (2) 64 MHz |
| (3) 99 MHz | (4) 89 MHz |

Space for your notes:

Q10 - 01 February - Shift 2

In an amplitude modulation, a modulating signal having amplitude of X V is superimposed with a carrier signal of amplitude Y V in first case. Then, in second case, the same modulating signal is superimposed with different carrier signal of amplitude $2Y$ V. The ratio of modulation index in the two case respectively will be:

- (1) 1 : 2 (2) 1 : 1
(3) 2 : 1 (4) 4 : 1

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (4)

Q2 (1)

Q3 (3)

Q4 (4)

Q5 (2)

Q6 (4)

Q7 (2)

Q8 (4)

Q9 (2)

Q10 (3)

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Q1 (4)

Modulation index

$$\mu = \frac{\text{Amplitude of modulating signal}}{\text{Amplitude of carrier wave}}$$

$$\mu = \frac{1}{2}$$

Q2 (1)

AM Broadcast $\rightarrow$ 540 – 1600 KHz $\rightarrow$ FM Broadcast $\rightarrow$ 88 – 108 MHzTelevision $\rightarrow$ 54 – 890 MHzSatellite communication $\rightarrow$ 3.7 – 4.2 GHz $\therefore$ A-II, B-I, C-IV, D-III

Q3 (3)

Given

Signal frequency $f_m = 5\text{kHz}$

Carrier wave frequency $f_c = 2\text{MHz}$

$f_c = 2000\text{KHz}$

The resultant signal will have band width of frequency given by

$$[(f_c + f_m) - (f_c - f_m)]$$

$$\Rightarrow [(2000 + 5) - (2000 - 5)]\text{kHz}$$

$$\Rightarrow 10\text{ kHz}$$

Q4 (4)

Maximum line of sight distance between two

antennas, $d_M = \sqrt{2Rh_T} + \sqrt{2R.h_R}$

Q5 (2)

$$\mu = \text{modulation index} = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

$$= \frac{14 - 6}{14 + 6} = 0.4$$

Q6 (4)

Solution not Available

Q7 (2)

Carrier wave frequency

$$V_c = \frac{100\pi}{2\pi} = 500 \text{ Hz}$$

Modulating wave frequency

$$V_m = \frac{4\pi}{2\pi} = 2 \text{ Hz}$$

$$\therefore V_c - V_m, V_c, V_c + V_m$$

$$= 498 \text{ Hz}, 500 \text{ Hz}, 502 \text{ Hz}$$

Q8 (4)

No Solution

Q9 (2)

FM broadcast range is 88MHz to 108MHz

Q10 (3)

Modulating Index

$$\mu = \frac{A_m}{A_c}$$

$$\mu_1 = \frac{X}{Y}$$

$$\mu_2 = \frac{X}{2Y}$$

$$\frac{\mu_1}{\mu_2} = \frac{2}{1}$$

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